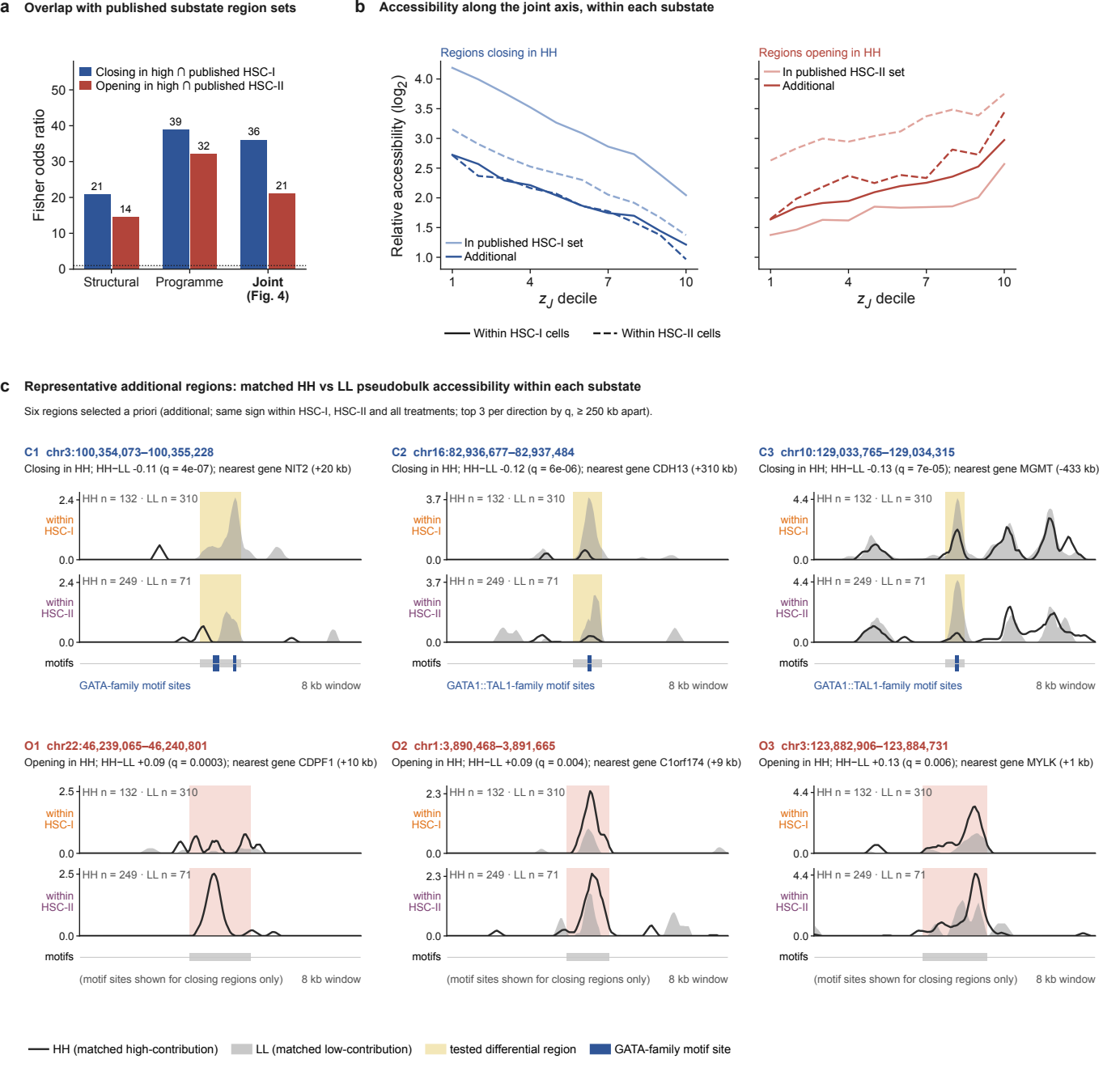

a Overlap with published substate region sets
Closing in high ∩ published HSC-I
Opening in high ∩ published HSC-II
Fisher odds ratio
50
40
30
20
10
0
21
14
39
32
36
21
Structural
Programme
Joint
(Fig. 4)
b Accessibility along the joint axis, within each substate
Regions closing in HH
Regions opening in HH
Relative accessibility (log2)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
In published HSC-I set
Additional
In published HSC-II set
Additional
1
4
7
10
zJ decile
Within HSC-I cells
Within HSC-II cells
c Representative additional regions: matched HH vs LL pseudobulk accessibility within each substate
Six regions selected a priori (additional; same sign within HSC-I, HSC-II and all treatments; top 3 per direction by q, ≥ 250 kb apart).
C1 chr3:100,354,073–100,355,228
Closing in HH; HH−LL -0.11 (q = 4e-07); nearest gene NIT2 (+20 kb)
C2 chr16:82,936,677–82,937,484
Closing in HH; HH−LL -0.12 (q = 6e-06); nearest gene CDH13 (+310 kb)
C3 chr10:129,033,765–129,034,315
Closing in HH; HH−LL -0.13 (q = 7e-05); nearest gene MGMT (-433 kb)
HH n = 132 · LL n = 310
HH n = 249 · LL n = 71
within HSC-I
within HSC-II
motifs
GATA-family motif sites
GATA1::TAL1-family motif sites
8 kb window
O1 chr22:46,239,065–46,240,801
Opening in HH; HH−LL +0.09 (q = 0.0003); nearest gene CDPF1 (+10 kb)
O2 chr1:3,890,468–3,891,665
Opening in HH; HH−LL +0.09 (q = 0.004); nearest gene C1orf174 (+9 kb)
O3 chr3:123,882,906–123,884,731
Opening in HH; HH−LL +0.13 (q = 0.006); nearest gene MYLK (+1 kb)
(motif sites shown for closing regions only)
HH (matched high-contribution)
LL (matched low-contribution)
tested differential region
GATA-family motif site